Pre-analysis plan | RT

In this pre-analysis plan, we detail how we will test each of the four causal hypotheses (H1-H4) of the #ManyDaughters project. For each hypothesis, we describe the variables, the inclusion and exclusion criteria for observations, the exact specification of the regression model, and the interpretation of results, i.e., the statistic used for testing the hypothesis. We will use data from the SOEP-Core for all analyses.

Testing Hypothesis 1

H1. *Having daughters leads to stronger support for gender equality (as measured by attitudes).*
That is, denoting X as a measure of support for gender equality,
(X for respondents with daughters) > (X for respondents without daughters).

We will test H1 using a fixed effects regression model. We will predict respondents' attitudes toward gender equality as a function of whether a given respondent had one or multiple daughters, or else no children or only sons (measured as a binary variable). The fixed effects model will take into account unobserved, time-invariant characteristics of the respondents that might otherwise confound the association between the predictor variable (having (a) daughter(s)) and the outcome (respondents' attitudes toward gender equality).

Variables

Outcome
A respondent's attitude toward gender equality will be measured using the following items:

- **f12p116k:** Fam: Man/Women Equal in Household/Family and Job (wave 2012)
- **plh0308_v2:** Best if man and woman work the same amount so they can share the responsibility (wave 2018)
- **lb1120:** Fam: Man/Women Equal in Household/Family and Job (wave 2020)

Unfortunately, we were unable to find an explicit description of the wording of the survey items f12p116k and lb1120 in the SOEP online companion or the questionnaires. We assume that these items essentially capture respondents' attitude toward gender equality similarly like the items plh0308_v2. Note that we only identified three items that seem to measure attitudes toward gender equality. These items were collected in panel waves 2012, 2018, and 2020. We discussed whether it would be appropriate or necessary to substitute or impute data for the waves that did not include the item. However, we decided against such a practice because it would be highly speculative. However, we believe that the three measurement time points will still allow us to test H1 given that we will use a fixed effects model that only examines within-person variation over time. It will obviously be important to acknowledge the limitations resulting from this approach (few observed waves with the outcome, i.e., small sample size, and less precise coefficient estimates).

Predictor

We will use a binary variable to measure whether a given respondent has (1) one or multiple daughters or (0) no children or one or multiple sons. We will identify when the first and subsequent daughters were born, using the variable **kidgeb01** – kidgeb*n (SOEP-Core V37 – BIOBIRTH), which measures the year of birth of a child, and the variable **kidsex01** – kidsex*n (SOEP-Core V37 – BIOBIRTH), which measures the sex of the first to the n-th child and allows us to distinguish male from female children. We assume that the event of having one daughter will cause most of the change in the respondent's attitude toward gender equality, if at all. While having more than one daughter might be associated with incremental change in respondents' attitudes, we expect the fundamental change in attitudes to be caused by the fact that a daughter raises the respondent's awareness of the importance of gender equality, if there is any change. This binary variable will be assessed at all panel waves observed here (from 2000 to 2020), allowing us to assess the effect of a within-respondent change on the outcome, albeit only in respondents who became the parents of a daughter in the course of the observation period. For other respondents, we will be unable to gauge such an effect as there will be no pertinent within-respondent change.

Covariates

We will estimate fixed effects regression models to analyze the effect of within-person change on the outcome. In such models, we do not control for time-invariant potential confounders because fixed effects models eliminate the effect of any variables that do not change over time. However, we will include a set of time-varying covariates to control for potential confounding. In particular, we will control for *age, number of children in the household,* and *partnership status*.

We will consider *age* because it represents aging effects and serves as a proxy for developmental stages which can influence attitudes regardless of whether a respondent has a daughter. The age variable will be constructed as follows: Age in the year 2000 will be computed as 2000-**gebjahr** (SOEP-Core v39 – PPATHL: Person-Related Meta-Dataset). Age in the following years will be computed as (2000-**gebjahr** + *t*), where *t* takes on the values 1-20 for the waves 2001-2020.

We will consider the number of children in the household because the effect of having a daughter might be minimized by the presence of siblings. We will use the **sumkids** variable to control for the number of children in the household at a given panel wave (SOEP-Core V37 – BIOBIRTH data).

We will consider *partnership status* because changes in the partnership status might constitute an alternative explanation for changes in attitudes toward gender equality given that a partner might actively negotiate matters relating to gender inequality with the respondent and thus raise awareness for the importance of—or undermine support for—gender equality. We will use the **pgpartz** variable from the *pgen* dataset to assess the respondent's partnership status, distinguishing between (0) no partner and (1) spouse/registered partner, partner, probably spouse/registered partner, probably partner.

Inclusion/exclusion criteria

We will exclude individuals who already have daughters prior to the year 2000. This is because we only focus on individuals who experience a change in the predictor during the observation window. We will include data from as many waves as possible.

Exact analysis

We will test H1 using a fixed effects regression model. We will predict respondents' attitudes toward gender equality as a function of whether they had one or multiple daughters in the preceding panel wave (as opposed to no children or only sons). We will focus on within-respondent changes only and specify the model as follows:

$$y_{it+1} = \beta_0 + \beta_1 x_{it} + \sum_{h=1}^{l} \beta_h x_{it} + \varepsilon_{it+1} \quad \text{(eq. 1)}$$

Here, the attitude toward gender equality y of a respondent i at time $t + 1$ is estimated as a function of the overall mean attitude over time (i.e., the intercept β_0), the main predictor x_{it}, that is, whether respondent i had one or multiple daughters at time t, a vector of time-varying control variables (x_{hit} to x_{lit}, specifically, age, number of children, and partnership status), and an error term that encompasses all unobserved random noise affecting the outcome at time $t + 1$ (ε_{it+1}), where we measure the lagged outcome. The error term is assumed to have a zero mean given the predictors, to be drawn from a normally distributed population, and to be independent of the predictors. We will estimate cluster-robust standard errors at the individual level (using the pid variable) to account for autocorrelation within individuals, thereby adjusting for dependence within each individual's error terms over time, which is common in longitudinal data. We will employ weights to adjust the estimation to attrition bias, using the inverse staying probability weight (pbleib). Moreover, we will employ weights to correct for differential sampling probabilities, using the inverse sampling probability weight (design), if this variable is available in the datasets we have access to.

Interpretation of results

We will interpret the t-statistic, examining whether it differs from zero, along with the corresponding p-value, to test whether having daughters affects attitudes toward gender equality. We will consider a p-value < .05 as evidence in line with the hypothesis that having daughters leads to stronger support for gender equality.

Testing Hypothesis 2

H2. *Having daughters leads to more left-leaning political preferences.*
That is, denoting Y as a measure of left-leaning political preferences,
(Y for respondents with daughters) > (Y for respondents without daughters).

We will test H2 using a fixed effects regression model. We will predict respondents' left-leaning political preferences as a function of whether a given respondent had one or multiple daughters, or else no children or only sons. The fixed effects model will take into account unobserved, time-invariant characteristics of the respondents that might otherwise confound the association between the predictor and the outcome.

Variables

Outcome
A respondent's political preferences will be measured using the **plh0004** from the *pl* dataset. Respondents were asked to answer the following question on a scale from 0 (entirely on the left) to 10 (entirely on the right): «In politics people often talk about "left" and "right" when it comes to characterize different political attitudes. If you think about your own political views: Where would you place yours?». This item was assessed in the following waves: 2005, 2009, 2014, and 2019. We will use data from all these waves in our model. Note that we will reverse code the item such that 10 will indicate left-leaning political preferences and 0 will represent right-leaning political preferences.

Predictor
As in the model testing H1, we will use a binary variable to measure whether a given respondent has (1) one or multiple daughters or (0) no children or one or multiple sons. This binary variable will be assessed at all panel waves observed here (from 2000 to 2020), allowing us to assess the effect of a within-respondent change on the outcome, albeit only in respondents who became the parents of a daughter in the course of the observation period. For other respondents, we will be unable to gauge such an effect as there will be no pertinent within-respondent change. For details on the operationalization, see above.

Covariates
As in the model testing H1, we will estimate fixed effects regression models to analyze the effect of within-person change on the outcome. In such models, we do not control for time-invariant potential confounders because fixed effects models eliminate the effect of any variables that do not change over time. However, we will include *age, number of children in the household,* and *partnership status* as time-varying covariates in the model (see above for details on the rationale and the operationalizations of these variables).

Inclusion/exclusion criteria

We will exclude individuals who already have daughters prior to the year 2000, because we only focus on individuals who experience a change in the relevant predictor during the observation window. We will include data from as many waves as possible.

Exact analysis

We will test H2 using a fixed effects regression model. We will predict respondents' left-leaning political preferences as a function of whether a given respondent had one or multiple daughters (as opposed to either no children or only sons). We will focus on within-respondent changes only and specify the model as follows:

$$y_{it+1} = \beta_0 + \beta_1 x_{it} + \sum_{h=1}^{l} \beta_h x_{it} + \varepsilon_{it+1} \quad \text{(eq. 2)}$$

Here, the political preference y of a respondent i at time $t + 1$ is estimated as a function of the overall mean political preference over time (i.e., the intercept β_0), the main predictor x_{it}, that is, whether respondent i has one or multiple daughters at time t, a vector of time-varying control variables (x_{hit} to x_{lit}, specifically, specifically, age, number of children, and partnership status), and an error term that encompasses all unobserved random noise affecting the outcome at time $t + 1$ (ε_{it+1}). The error term is assumed to have a zero mean given the predictors, to be drawn from a normally distributed population, and to be independent of the predictors. We will estimate cluster-robust standard errors at the individual level (using the pid variable) to account for autocorrelation within individuals, thereby adjusting for dependence within each individual's error terms over time, which is common in longitudinal data. We will employ weights to adjust the estimation to attrition bias, using the inverse staying probability weight (pbleib). We will also employ weights to correct for differential sampling probabilities, using the inverse sampling probability weight (design), if this variable is available in the datasets we have access to.

Interpretation of results

We will interpret the t-statistic, examining whether it differs from zero, along with the corresponding p-value, to test whether having daughters affects attitudes toward gender equality. We will consider a p-value < .05 as evidence in line with the hypothesis that having daughters leads to more left-leaning political preferences.

Testing Hypothesis 3

H3. *Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).*
That is, denoting Z as a measure of gender equality in the household,
(Z for respondents with daughters) > (Z for respondents without daughters).

We will test H3 using a fixed effects regression model. We will predict households' gender equality as a function of whether a given household has one or multiple daughters, or else no children or only sons. The fixed effects model will take into account unobserved, time-invariant characteristics of the household that might otherwise confound the association between the predictor and the outcome.

Variables

Outcome
A households' gender equality will be measured using an index of unequal participation in the labor market. In particular, we will investigate the difference in the number of working hours between partners ($\Delta X = X_{\text{male respondent's working hours}} - X_{\text{female respondent's working hours}}$), ultimately evaluating whether this difference changes as a result of the birth of a daughter. To this end, we will rely on the variable **plb0176_h** which measures the agreed weekly working hours (harmonized). The variable is available (in the *pl* dataset) from 2000 to 2019. We will use the 2019 value to substitute the missing 2020 variable. To construct the difference score measuring inequality among the dyad, we will use the person ID (pid), the current wave household ID (hid), and if needed, the case-id (cid), that is, the original household number, alongside the sex variable which will allow us to determine the respondent's sex. For the construction of this variable, we will assume that partners are mainly heterosexual. However, the operationalization also allows us to measure unequal participation in the labor market among same-sex partners. For them, the outcome will measure inequality among the partners. We believe that it is appropriate not to exclude same-sex partners from the analysis.

Predictor
We will use a binary variable to measure whether a given household has (1) one or multiple daughters or else (0) no children or only sons. This variable will be assessed at all panel waves observed (2000-2012), allowing us to assess the effect of a within-household change on the outcome, albeit only in households that gained a daughter in the course of the observation period. For details on the operationalization, see above. For details on the operationalization, see above.

Covariates
As covariate, we will take into account the number of kids in the household in a given wave, using the **sumkids** variable from SOEP-Core V37 – BIOBIRTH.

Inclusion/exclusion criteria

We will exclude households that already have daughters prior to the year 2000. This is because we can only focus on individuals who experience a change in the relevant predictor during the observation window. We will include data from as many waves as possible. We will also exclude households that experience a dissolution of the partnership.

Exact analysis

We will test H3 using a fixed effects regression model. We will predict a households' gender equality as a function of whether it had one or multiple daughters in the preceding panel wave(s) (as opposed to no children or only sons). We will focus on within-household changes only and specify the model as follows:

$$y_{it+1} = \beta_0 + \beta_1 x_{it} + \sum_{h=1}^{l} \beta_h x_{it} + \varepsilon_{it+1} \quad \text{(eq. 3)}$$

Here, the gender equality y of a household i at time $t + 1$ is estimated as a function of the overall mean gender equality over time (i.e., the intercept β_0), the main predictor x_{it}, that is, whether household i has one or multiple daughters at time t, a vector of time-varying control variables (x_{hit} to x_{lit}, specifically, mean age of parents in the household and number of children in the household), and an error term that encompasses all unobserved random noise affecting the outcome at time $t + 1$ (ε_{it+1}). The error term is assumed to have a zero mean given the predictors, to be drawn from a normally distributed population, and to be independent of the predictors. We will estimate cluster-robust standard errors at the household level (using the hid variable) to account for autocorrelation within households.

Interpretation of results

We will interpret the *t*-statistic, examining whether it differs from zero, along with the corresponding *p*-value, to test the hypothesis that having daughters leads to more gender equality in the household. We will consider a *p*-value < .05 as evidence in line with the hypothesis.

Testing Hypothesis 4

H4. *Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).*
That is, denoting X as a measure of support for gender equality,
(X for male respondents with daughters − X for male respondents without daughters) > (X for female respondents with daughters − X for female respondents without daughters).

We will test H4 using a fixed effects regression model with an interaction term for respondents' sex. We will predict respondents' attitudes toward gender equality as a function of whether a given respondent has one or multiple daughters, or no children or only sons, and will include an interaction term between the respondent's sex and the time-varying variable "having a daughter" to assess differential effects for fathers and mothers (or male and female guardians).

Variables

Outcome
A respondent's attitude toward gender equality will be measured using the same indicator as we use in the model testing H1 (see above).

Predictors

As in the other models, we will use a binary variable to measure whether a given respondent has (1) one or multiple daughters or (0) no children or one or multiple sons. This binary variable will be assessed at all panel waves observed here (from 2000 to 2020), allowing us to assess the effect of a within-respondent change on the outcome, albeit only in respondents who became the parents of a daughter in the course of the observation period. For other respondents, we will be unable to gauge such an effect as there will be no pertinent within-respondent change. For details on the operationalization, see above.

Additionally, the second key predictor is a respondent's sex (male vs. female). We will use the **sex** variable from the SOEP-Core v39 – PPATHL: Person-Related Meta-Dataset.

Covariates

As in the other models, we will estimate fixed effects regression models to analyze the effect of within-person change on the outcome. In such models, we do not control for time-invariant potential confounders because fixed effects models eliminate the effect of any variables that do not change over time. However, we will include *age, number of children in the household,* and *partnership status* as time-varying covariates in the model (see above for details on the rationale and the operationalizations of these variables).

Inclusion/exclusion criteria

We will exclude individuals who already have daughters prior to the year 2000. This is because we only focus on individuals who experience a change in the relevant predictor during the observation window. We will include data from as many waves as possible.

Exact analysis

We will test H4 using a fixed effects model. We will predict respondents' attitudes toward gender equality as a function of whether a given respondent had one or multiple daughters in the preceding panel wave(s) (as opposed to either no children or only sons). We will focus on within-respondent changes only and specify the model as follows:

$$y_{it+1} = \beta_0 + \beta_1 x_{it} + \beta_2 (x_{it} \times M_i) + \sum_{h=1}^{l} \beta_h x_{it} + \varepsilon_{it+1} \quad \text{(eq. 4)}$$

Here, the attitude toward gender equality y of a respondent i at time $t + 1$ is estimated as a function of the overall mean attitude over time (i.e., the intercept β_0), the main predictor x_{it}, that is, whether respondent i has one or multiple daughters at time t, an interaction between the main predictor x_{it} and an indicator for being male M_i, a vector of time-varying control variables (x_{hit} to x_{lit}, specifically, age, number of children in the household, and partnership status), and an error term that encompasses all unobserved random noise affecting the outcome at time $t + 1$ (ε_{it+1}). The error term is assumed to have a zero mean given the predictors, to be drawn from a normally distributed population, and to be independent of the predictors. We will estimate cluster-robust standard errors at the individual level to account for autocorrelation within individuals, thereby adjusting for dependence within each

individual's error terms over time, which is common in longitudinal data. We will employ weights to adjust the estimation to attrition bias, using the inverse staying probability weight (pbleib). Moreover, we will employ weights to correct for differential sampling probabilities, using the inverse sampling probability weight (design), if this variable is available in the datasets we have access to.Finally, note that the main effect of sex is dropped from the model, because it is constant within individuals. However, it will be used as part of the interaction term.

Interpretation of results

We will interpret the *t*-statistic for the interaction term, examining whether it differs from zero, along with the corresponding *p*-value ($p < .05$ level) as evidence in line with the hypothesis that having daughters leads to stronger support for gender equality among men than among women.

Summary of all the variables used in the analyses

Person ID
pid (*SOEP-Core v39 – PPATHL: Person-Related Meta-Dataset*)

household ID
hid *(SOEP-Core v39 – PPATHL: Person-Related Meta-Dataset*)
cid (alternative measure)

Survey year
syear (*SOEP-Core v39 – PPATHL: Person-Related Meta-Dataset*)

Respondent's age
gebjahr (*SOEP-Core v39 – PPATHL: Person-Related Meta-Dataset*)

Respondent's sex
sex (*SOEP-Core v39 – PPATHL: Person-Related Meta-Dataset*)

Year of birth of a child
kidgeb01 (SOEP-Core V37 – BIOBIRTH)

Sex of the child
kidsex01 (SOEP-Core V37 – BIOBIRTH)

Number of kids in the household in a given wave
sumkids (SOEP-Core V37 – BIOBIRTH)

Left-leaning political preferences
plh0004 (pl dataset, assessed in 2005, 2009, 2014, 2019)

Partnership status
pgpartz (pgen dataset) or **partner** (ppathl dataset)

Attitudes toward gender equality
f12p116k: Fam: Man/Women Equal in Household/Family and Job
plh0308_v2: Best if man and woman work the same amount so they can share the responsibility
lb1120: Fam: Man/Women Equal in Household/Family and Job

Gender equality as measured by behaviors or outcomes
pl/plb0176_h: Agreed weekly working hours (pl dataset, variable available from 2000 to 2019)

Weights
pbleib (inverse staying probability)
design (inverse sampling probability; if the datasets available to us include this variable)